MATH 181A - Homework 5

due Thursday, February 12 at 11:59PM

Problem 1.

Let $X_1, \ldots, X_n$ be i.i.d. random variables from a Geometric(p), $p \in (0,1)$, which means that $P_X(k;p) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$ and we have $\mathbb{E}[X_i] = 1/p$.

1.1 Find the asymptotic variance of the maximum likelihood estimator $\hat{p}$.

Solution: The MLE is found as

$$\ell(p) = \sum_{i=1}^{n} \log\left((1-p)^{x_i-1}p\right) = \sum_{i=1}^{n}(x_i - 1)\log(1-p) + \sum_{i=1}^{n}\log p$$

$$\frac{d}{dp}\ell(p) = -\sum_{i=1}^{n}\frac{x_i - 1}{1-p} + \sum_{i=1}^{n}\frac{1}{p} = \frac{n}{1-p} - \frac{1}{1-p}\sum_{i=1}^{n}x_i + \frac{n}{p} = 0$$

$$\frac{n}{1-p} - \frac{1}{1-p}\sum_{i=1}^{n}x_i + \frac{n}{p} = 0 \implies n - \sum_{i=1}^{n}x_i + \frac{n}{p} - n = 0 \implies \sum_{i=1}^{n}x_i = \frac{n}{p}$$

$$\hat{p} = \frac{n}{\sum_{i=1}^{n}x_i} = \frac{1}{\bar{X}}$$

The Fisher information is:

$$\begin{aligned} I(p) &= -\mathbb{E}\left[\frac{d^2}{dp^2}\log\left((1-p)^{X-1}p\right)\right] \\ &= -\mathbb{E}\left[\frac{d^2}{dp^2}(X-1)\log(1-p) + \log p\right] \\ &= \mathbb{E}\left[\frac{(X-1)}{(1-p)^2} + \frac{1}{p^2}\right] = \frac{1-p}{p(1-p)^2} + \frac{1}{p^2} = \frac{1}{p^2(1-p)} \end{aligned}$$

Thus, the asymptotic variance of the MLE is:

$$\mathrm{Var}(\hat{p}) = \frac{1}{nI(\hat{p})} = \frac{p^2(1-p)}{n}$$

1.2 Let $n = 100$ and you observe the following data:

Number	1	2	3	4	5
Frequency	60	21	12	4	3

which means that 60 of the random variables $X_1, \ldots, X_{100}$ equal 1, 21 of them equal 2, and so on. Find a 95% confidence interval for p based on the MLE.

Solution: The MLE for p is $\hat{p} = 1/\bar{X} = 1/1.69 \approx 0.5917$. The 95% confidence interval for p is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{1}{nI(\hat{p})}} = 0.5917 \pm 1.96\sqrt{\frac{(0.5917)^2(1-0.5917)}{100}} = (0.5176, 0.6658)$$

Problem 2.

Let $X_1, \ldots, X_n$ be i.i.d. random variables from a $N(\sqrt{\mu}, 1)$ for $\mu \geq 0$. The MLE for μ is $\hat{\mu} = (\bar{X})^2$.

2.1 Find the (exact) variance of the MLE $\hat{\mu}$. Hint: You may use the fact that if $Y \sim N(\mu, \sigma^2)$, $\mathbb{E}[(Y-\mu)^4] = 3\sigma^4$.

Solution:

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\mu} = \sqrt{\mu}$$

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n} 1 = \frac{1}{n}$$

$$\mathrm{Var}(\hat{\mu}) = \mathrm{Var}((\bar{X})^2) = \mathbb{E}[(\bar{X})^4] - (\mathbb{E}[(\bar{X})^2])^2$$

$$\mathbb{E}[(\bar{X})^2] = \mathrm{Var}(\bar{X}) + (\mathbb{E}[\bar{X}])^2 = \frac{1}{n} + (\sqrt{\mu})^2 = \frac{1}{n} + \mu$$

Let $\bar{Z} \sim N(0, \frac{1}{n})$ then:

$$\mathbb{E}[(\bar{X})^4] = \mathbb{E}[(\bar{Z} + \sqrt{\mu})^4] = \mathbb{E}[\bar{Z}^4 + 4\bar{Z}^3\sqrt{\mu} + 6\bar{Z}^2\mu + 4\bar{Z}\mu^{3/2} + \mu^2] = \frac{3}{n^2} + \frac{6\mu}{n} + \mu^2$$

$$\mathrm{Var}(\hat{\mu}) = \mathbb{E}[(\bar{X})^4] - (\mathbb{E}[(\bar{X})^2])^2 = \frac{3}{n^2} + \frac{6\mu}{n} + \mu^2 - \left(\frac{1}{n^2} + 2\frac{\mu}{n} + \mu^2\right) = \frac{2}{n^2} + \frac{4\mu}{n}$$

2.2 Find the asymptotic variance of the MLE $\hat{\mu}$.

Solution: The Fisher information is:

$$I(\mu) = -\mathbb{E}\left[\frac{d^2}{d\mu^2}\log f_X(x;\mu)\right] = -\mathbb{E}\left[\frac{d}{d\mu}\frac{1}{2\sqrt{\mu}}(x - \sqrt{\mu})\right] = \mathbb{E}\left[\frac{1}{4\mu} + \frac{1}{4\sqrt{\mu^3}}(x - \sqrt{\mu})\right] = \frac{1}{4\mu}$$

Thus, the asymptotic variance of the MLE is:

$$\mathrm{Var}(\hat{\mu}) = \frac{1}{nI(\hat{\mu})} = \frac{4\mu}{n}$$

Problem 3. 5.7.2

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n from a normal pdf having $\mu = 0$. Show that $S_n^2 = \frac{1}{n}\sum_{i=1}^n Y_i^2$ is a consistent estimator for $\sigma^2 = \text{Var}(Y)$. Hint: Likewise, you may use the fact that if $Y \sim N(0, \sigma^2)$, $\mathbb{E}(Y^4) = 3\sigma^4$.

Solution:

$$\mathbb{E}[S_n^2] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n Y_i^2\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[Y_i^2] = \frac{1}{n}\sum_{i=1}^n \text{Var}(Y_i) + (\mathbb{E}[Y_i])^2 = \frac{1}{n}\sum_{i=1}^n \sigma^2 + 0 = \sigma^2 \implies \text{unbiased}$$

$$\text{Var}(S_n^2) = \text{Var}\left[\frac{1}{n}\sum_{i=1}^n Y_i^2\right] = \frac{1}{n^2}\text{Var}\left[\sum_{i=1}^n Y_i^2\right] = \frac{1}{n^2}\sum_{i=1}^n \text{Var}(Y_i^2)$$

$$= \frac{1}{n^2}\sum_{i=1}^n \mathbb{E}[Y_i^4] - \left(\mathbb{E}[Y_i^2]\right)^2 = \frac{1}{n^2}\sum_{i=1}^n 3\sigma^4 - \sigma^4 = \frac{2\sigma^4}{n}$$

Since $\lim_{n\to\infty} \text{Var}(S_n^2) = 0$ then S_n^2 is a consistent estimator for σ^2.

Problem 4.

Let $Y_1, Y_2, \ldots, Y_n$ be a random sample of size n from the pdf $f_Y(y;\theta) = \frac{1}{\theta}e^{-y/\theta}, y > 0$. Let $\hat{\theta}_1 = Y_1$, $\hat{\theta}_2 = \bar{Y}$, and $\hat{\theta}_3 = n \cdot Y_{\min}$ Show if each of the estimators is consistent for θ or not. You can use some of the results you have done in Homework 4 Problem 1.

Solution: We know from HW 4 1.1 that they are all unbiased estimators for θ. We also know the variances from HW 4 1.2 with $\text{Var}(\hat{\theta}_1) = \text{Var}(\hat{\theta}_3) = \theta^2$, $\text{Var}(\hat{\theta}_2) = \theta^2/n$. Hence the only estimator that is consistent for θ is $\hat{\theta}_2$ since $\lim_{n\to\infty} \text{Var}(\hat{\theta}_2) = 0$ while $\lim_{n\to\infty} \text{Var}(\hat{\theta}_1) = \lim_{n\to\infty} \text{Var}(\hat{\theta}_3) = \theta^2 \neq 0$.

Problem 5.

Suppose $\hat{\theta}_n = Y_{\max}$ is to be used as an estimator for the parameter θ in the uniform pdf, $f_Y(y;\theta) = 1/\theta, 0 \leq y \leq \theta$. Show that $\hat{\theta}_n$ is squared-error consistent (see Question 5.7.4).

Solution: $\hat{\theta}_n$ is squared-error consistent if $\lim_{n\to\infty} \mathbb{E}[(\hat{\theta}_n - \theta)^2] = 0$. First, we'll need the first and second moments of $Y_{\max}$.

$$\mathbb{P}(Y_{\max} \leq y) = \mathbb{P}(Y_1 \leq y, \ldots, Y_n \leq y) = \mathbb{P}(Y_1 \leq y)^n = \left(\frac{y}{\theta}\right)^n$$

$$f_{Y_{\max}}(y) = \frac{d}{dy}\mathbb{P}(Y_{\max} \leq y) = \frac{d}{dy}\left(\frac{y}{\theta}\right)^n = n\left(\frac{y}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} = \frac{n}{\theta}\left(\frac{y}{\theta}\right)^{n-1}$$

$$\mathbb{E}[Y_{\max}] = \int_0^\theta y \frac{n}{\theta}\left(\frac{y}{\theta}\right)^{n-1} dy = \frac{n}{\theta^n}\int_0^\theta y^n dy = \frac{n}{\theta^n}\left[\frac{y^{n+1}}{n+1}\right]_0^\theta = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n\theta}{n+1}$$

$$\mathbb{E}[Y_{\max}^2] = \int_0^\theta y^2 \frac{n}{\theta}\left(\frac{y}{\theta}\right)^{n-1} dy = \frac{n}{\theta^n}\int_0^\theta y^{n+1} dy = \frac{n}{\theta^n}\left[\frac{y^{n+2}}{n+2}\right]_0^\theta = \frac{n}{\theta^n} \cdot \frac{\theta^{n+2}}{n+2} = \frac{n\theta^2}{n+2}$$

$$\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \mathbb{E}[(Y_{\max} - \theta)^2] = \mathbb{E}[Y_{\max}^2] - 2\theta\mathbb{E}[Y_{\max}] + \theta^2 = \frac{n\theta^2}{n+2} - \frac{2n\theta^2}{n+1} + \theta^2 = \frac{2\theta^2}{n^2+3n+2}$$

Since $\lim_{n\to\infty} \mathbb{E}[(\hat{\theta}_n - \theta)^2] = 0$ then $\hat{\theta}_n$ is squared-error consistent.

Problem 6.

Suppose $X_1, X_2, \ldots$ is a sequence of i.i.d. random variables having the Poisson distribution with mean λ. Let $\hat{\lambda}_n = X_n$ (the last random variable).

6.1 Is $\hat{\lambda}_n$ an unbiased estimator of λ? Explain your answer.

Solution: Yes, since $\mathbb{E}[\hat{\lambda}_n] = \mathbb{E}[X_n] = \mathbb{E}[X_i] = \lambda$.

6.2 Is $\hat{\lambda}_n$ a consistent estimator of λ? Explain your answer.

Solution: No, since $\mathrm{Var}(\hat{\lambda}_n) = \mathrm{Var}(X_n) = \mathrm{Var}(X_i) = \lambda$. Hence $\lim_{n\to\infty} \mathrm{Var}(\hat{\lambda}_n) = \lambda \neq 0$.

Problem 7. R Simulation

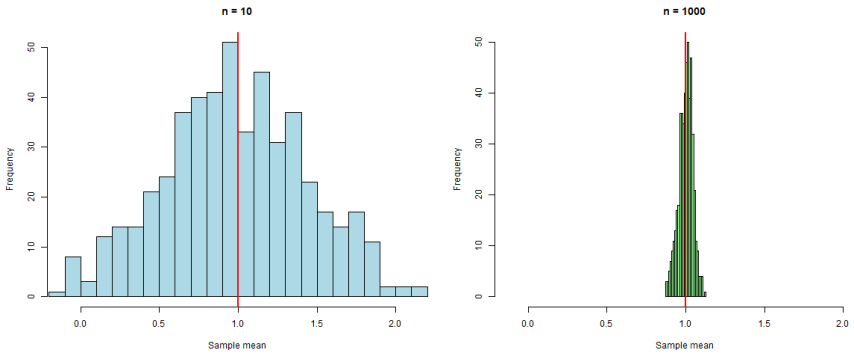
7.1 For $n = 1000$, simulate a random sample of size n from $N(0, 1)$. Use the generated data to give an approximation to the critical values when $\alpha = 0.01, 0.05, 0.1$, and compare them with the theoretical values $z_{\alpha/2}$. Repeat with $n = 10,000$ to get a better approximation. Hint: Commands `quantile()`, `qnorm()` may be helpful.

```
[1] "Theoretical z_{alpha/2}:"
alpha=0.01 alpha=0.05  alpha=0.1
    2.5758     1.9600     1.6449
[1] "Approximation (n = 1000):"
alpha=0.01 alpha=0.05  alpha=0.1
    2.7020     1.8777     1.5335
[1] "Difference from theoretical:"
alpha=0.01 alpha=0.05  alpha=0.1
    0.1262    -0.0823    -0.1114
[1] "Approximation (n = 10000):"
alpha=0.01 alpha=0.05  alpha=0.1
    2.5533     1.9611     1.6619
[1] "Difference from theoretical:"
alpha=0.01 alpha=0.05  alpha=0.1
   -0.0225     0.0011     0.0171
```

Solution: The differences are about an order of magnitude smaller for $n = 10000$ compared to $n = 1000$ which is expected since more data is being used.

7.2 For $n = 10$, simulate a random sample of size n from $N(\mu, \sigma^2)$, where $\mu = 1$ and $\sigma^2 = 2$; compute the sample mean. Repeat the above simulation 500 times, plot the histogram of the 500 sample means. Now repeat the 500 simulations for $n = 1,000$. Compare these two sets of results with different sample sizes, and discuss it in the context of consistency.

Solution:



The $n = 1000$ histogram is much more concentrated around the mean of 1, while the $n = 10$ histogram is more spread out. This is expected since the variance of the sample mean is $\frac{\sigma^2}{n}$ and the variance of the sample mean of $n = 1000$ is much smaller than the variance of the sample mean of $n = 10$.

We know that the sample mean is a consistent estimator for the mean of the distribution, so as n increases, the sample mean will converge to the true mean. Hence, the histogram of the sample means will be more concentrated around the true mean.